

Toward a Bridge Between Theory and the Lab: LLM Agents in Bank-Run Games

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Abstract

This paper develops an experimental protocol that uses large language model agents as subjects in a standard bank-run coordination game. Each choice is generated independently, which allows us to rerun identical prompt histories while varying the information provided to the decision-makers. In a private-signal condition, choices are approximately threshold-based; reported beliefs about fundamentals track the Bayesian posterior mean for interior signals; and reasoning-token usage is highest near the empirical switching region. We then map an empirical best-response surface by experimentally fixing the number of other withdrawals presented as certain information. A payoff-irrelevant cue then shifts choices toward withdrawal and is associated with lower compliance with the imposed coordination. The approach offers a controlled complement to human-subject experiments for disentangling information and coordination forces in bank-run games.

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1 Introduction

Research on bank runs draws on both equilibrium theory and laboratory evidence. Canonical models, like [Diamond and Dybvig \(1983\)](#) and the global-games refinement of [Morris and Shin \(2003\)](#), deliver predictions for an unrepeated game under a specified information structure. Laboratory experiments, by contrast, typically collect repeated decisions from participants who may update beliefs and strategies across rounds. This difference may matter in coordination settings, where repetition can influence beliefs and strategies.

[Roth and Erev \(1995\)](#) show that behavior in extensive-form games does not reliably converge to equilibrium. Laboratory data are often shaped by transitional dynamics-learning, experimentation, and other adjustments induced by repetition. In the same spirit, [Fudenberg and Levine \(2016\)](#) argue that equilibrium is best viewed as a limiting benchmark of an adjustment process rather than a literal prediction for finite play. This creates a basic tension: statistical methodology requires repeated observations, yet repetition feeds back into behavior. Bank runs illustrate these tensions acutely. They are at the intersection of equilibrium multiplicity ([Diamond and Dybvig, 1983](#); [Morris and Shin, 2003](#)), informational externalities ([Treviño, 2020](#)), and experimentally documented path dependence that is difficult to decompose with human subjects ([Garratt and Keister, 2009](#); [Arifovic et al., 2013](#); [Kiss et al., 2012](#)). The experimental bank-run literature, surveyed in [Kiss et al. \(2022\)](#), identifies three broad mechanisms: fundamentals, coordination failure, and panic. Existing designs often trade off environmental richness against control and interpretability, making it difficult to study these mechanisms within a single, comparable framework.

This paper proposes one possible methodology. We use large language model (LLM) agents as controllable decision-makers in bank-run coordination games. LLM agents can be reset, re-run under identical informational histories, allowing tightly controlled manipulations of the information shown to the agent. We then evaluate whether this controllability yields interpretable best-response mappings in a standard bank-run game.

We run three conditions: (i) private signal only; (ii) private signal plus fixed others' action; (iii) the same with an added payoff-irrelevant cue. Experiment 1 validates the agent's

behavior against theoretical benchmarks: the withdrawal curve exhibits monotone threshold structure, reported beliefs track the Bayesian posterior, and deliberation effort (measured by the model’s reasoning-token usage) peaks near the switching region, a pattern documented in human response times (Heinemann et al., 2009). Experiment 2 maps the full best-response surface by exogenously fixing the number of other depositors who withdraw, separating the information channel (how the agent responds to its private signal about fundamentals) from the coordination channel (how it responds to others’ actions). The logit estimates suggest the coordination channel is at least as large as the information channel in this environment. Experiment 3 controls everything the same as Experiment 2, adds only a single payoff-irrelevant sentence, and measures the resulting shift in the surface. The cue reduces the aggregate rate and degrades the agent’s responsiveness to explicitly provided coordination information by roughly 80 percent. Rather than simply lowering the withdrawal threshold, the cue appears to dissolve the common-knowledge structure that the hard-conditioning block provides, pushing agents back toward a regime where the private signal is the dominant input.

Our protocol is to use LLM agents not as substitutes for human subjects, but as a complementary experimental tool that relaxes some of the constraints human experiments face. We do not claim that LLM agents solve the problems listed above. We do think they offer a different set of tradeoffs, and the bank-run coordination game is a setting where those tradeoffs may be useful. Instead, we use them for a narrow advantage that is valuable in coordination settings: we can run treatment and control information conditions, differing only in a single targeted intervention. A related property is that the micro-level decision process stays constant while the macro-structure varies. We treat the LLM agent specification as a maintained microfoundation, validate it against known human experimental regularities (Madiès, 2006; Schotter and Yorulmazer, 2009; Garratt and Keister, 2009; Arifovic et al., 2013), and then vary the informational and coordination environment so that outcome differences can be attributed to the environment rather than to shifting preferences or accumulated experience. A third property is that deliberation is partially observable. We

record model-generated explanations and token usage as process measures, while treating them as noisy proxies of cognition costs.

This paper attempts four things. First, we describe and implement a methodology for using LLM agents in coordination-game experiments, with emphasis on the identification strategies that the LLM setting makes feasible. We try to be explicit about what this methodology can and cannot do relative to theory and human experiments. Second, we build an experimental protocol for bank-run games that maps the joint role of fundamentals and strategic uncertainty across the best-response surface, and we use it to estimate the relative magnitude of the information and coordination channels. Third, we introduce a payoff-irrelevant framing cue and document its effect on the surface, providing evidence on a sunspot-style panic channel that is difficult to isolate cleanly in human experiments because of the confounding issues discussed above. Fourth, we document agent reasoning traces as auxiliary process data and report how they reason and proxy their deliberation levels. Several caveats deserve mention at the outset, and we discuss these issues in Section 5 and regard addressing them as natural next steps.

2 Experimental Design

We adopt a five-player depositor game in the spirit of [Bayona and Peia \(2022\)](#). The experiments share a common payoff environment and differ only in the information provided to each agent. Experiment 1 validates the agent against theoretical benchmarks and established human experimental regularities. Experiment 2 maps the two-dimensional best-response surface by independently varying fundamentals and the coordination environment. Experiment 3 introduces a single payoff-irrelevant cue and measures its effect on the surface. We run three designs that sequentially add (i) counterfactual information about others’ actions and (ii) an urgency cue, so we can compare behavior across nested information sets.

2.1 Common Environment

Five depositors simultaneously choose between two actions. Action A (withdraw) pays 100 points with certainty. Action B (stay) pays

$$\pi_B(\theta, n_A) = \frac{3\theta}{1 + 2n_A}, \quad (1)$$

where θ is the unobserved fundamental and $n_A \in \{0, 1, 2, 3, 4\}$ is the number of the other four depositors who withdraw. We adopt the payoff specification of [Bayona and Peia \(2022\)](#), who parameterize a finite-player global game based on [Goldstein and Pauzner \(2005\)](#). The payoff structure generates strategic complementarity: the return to staying falls as more others withdraw, and for any given n_A it is increasing in θ . When $n_A = 0$, staying dominates withdrawal for $\theta > 33$; when $n_A = 4$, staying dominates only for $\theta > 300$, which lies outside the support. The game therefore admits a wide intermediate region in which the optimal action depends on beliefs about both the fundamental and the behavior of others.

The fundamental θ is drawn uniformly from $[10, 150]$.¹ Each depositor i receives a private hint s_i drawn uniformly from $[\theta - 10, \theta + 10]$, clipped to $[10, 150]$. The signal structure (the distribution of θ , the noise width $\epsilon = 10$, and the fact that all depositors receive independent hints) is communicated to every agent as part of the prompt. Under this structure, the Bayesian posterior mean equals the signal ($E[\theta \mid s_i] = s_i$), so any deviation of the agent’s reported belief from its hint is a measurable departure from first-order rationality.

All agents are implemented using the same LLM (GPT-5.2) with temperature set to zero in experiments where the reasoning-effort parameter permits it. Each agent returns a structured JSON response that records the chosen action, a point estimate of θ , a point estimate of n_A , a categorical reason code, and a short free-text explanation. The structured-output format is enforced by the API, so every response is machine-parseable without post-

¹For the benchmark comparison $\mathbb{E}[\theta \mid s_i] = s_i$ to hold exactly under the stated finite-support model, we restrict the experimental design to interior realizations where boundary truncation is irrelevant. Concretely, we implement $\theta \in [20, 140]$, so the hint interval $[\theta - 10, \theta + 10]$ lies within $[10, 150]$ and no clipping occurs. Under a uniform prior on $[10, 150]$ and uniform noise width $\epsilon = 10$, this implies $\theta \mid s_i \sim \text{Unif}[s_i - 10, s_i + 10]$ and hence $\mathbb{E}[\theta \mid s_i] = s_i$.

hoc extraction. Full prompt texts and parameter settings are reported in Appendix A.

2.2 Experiment 1: Behavioral Validation

The first experiment places the agent in the simplest information environment that the theory considers. Each depositor observes only its private hint s_i and must decide whether to withdraw. No information about the actions of others is provided. This corresponds to the standard global-games setting in which the withdrawal decision is a function of the private signal alone: an agent withdraws if and only if its signal falls below some threshold s^* , and the equilibrium threshold is pinned down by the indifference condition at s^* (Carlsson and van Damme, 1993; Morris and Shin, 2003; Goldstein and Pauzner, 2005).

The experiment sweeps θ over the grid $\{20, 30, \dots, 140\}$, producing 13 grid points. At each grid point we run five independent seeds, each containing five agents who draw hints independently. This yields 25 decisions per value of θ and 325 decisions in total. The reasoning-effort parameter is set to “medium,” which allows the model to expend internal reasoning tokens before committing to an answer.

Experiment 1 serves two purposes. The first is calibration: we check whether the agent’s withdrawal curve exhibits the monotone threshold structure predicted by the theory, whether the empirical switching point is consistent with the break-even condition, and whether the agent’s reported beliefs track the Bayesian benchmark. The second is to establish process-level baselines. Because the LLM’s token usage on each decision is observable, we can measure whether deliberation effort varies with the difficulty of the decision, a pattern that has been documented in human response times in coordination-game experiments (Rubinstein, 2007; Heinemann et al., 2009). If the agent reproduces this regularity without having been trained on response-time data, it provides independent evidence that the decision rule responds to strategic ambiguity in a behaviorally plausible way.

2.3 Experiment 2: The Policy Surface

Experiment 1 confounds the information and coordination channels: the agent’s action reflects both what it infers about θ from its hint and what it expects others to do. Experiment 2 eliminates the second source of variation by telling each agent, with certainty, that exactly n_A^{cf} of the other four depositors will choose A. The prompt includes a hard-conditioning block (“You are 100% sure that exactly $\{n_A^{cf}\}$ of the OTHER investors will choose Action A. This is certain information, not a guess.”) that holds fixed the counterfactual $\{n_A^{cf}\}$, so variation across decisions reflects the hint about θ rather than beliefs about others, from the information channel.

The experiment varies θ over the coarser grid $\{20, 30, \dots, 140\}$ (13 values) and crosses it with five levels of $n_A^{cf} \in \{0, 1, 2, 3, 4\}$, producing a 13×5 grid of 65 cells. Each cell contains 25 independent decisions (five seeds, five agents per seed), for a total of 1,625 observations. Reasoning effort is set to “none” and temperature to zero, so that any behavioral variation within a cell comes from the stochastic hint draw rather than from model randomness.

The resulting object is the empirical best-response surface $\Pr(\text{Stay} \mid \theta, n_A^{cf})$. This surface is the structural primitive from which theoretical comparative statics are derived: equilibrium predictions correspond to the fixed points of the mapping under the consistency condition $n_A^{cf} = n_A^*(\theta)$. By estimating the surface directly, we can separate the information channel (how the withdrawal probability changes with θ holding n_A^{cf} fixed) from the coordination channel (how it changes with n_A^{cf} holding θ fixed). This separation is what makes the policy-surface approach useful: it recovers the two margins that theory distinguishes but that Experiment 1 (and most human experiments) observe only in combination.

A compliance check is built into the design. Each agent is asked to report its point estimate of n_A as part of the structured output. If the hard-conditioning is fully absorbed, every agent should report $\hat{n}_A = n_A^{cf}$. Any deviation indicates that the prompt has failed to override the agent’s endogenous coordination beliefs, a possibility that turns out to matter for interpreting Experiment 3.

2.4 Experiment 3: The Panic Channel

The final experiment tests whether a payoff-irrelevant framing cue can shift the best-response surface. The treatment is minimal: a single sentence (“Many other depositors may act quickly”) is appended to the system prompt, with other things keeping exact same as Experiment 2. This sentence does not change the payoff function, the distribution of θ or hints, or the grid of n_A^{cf} values. It is considered as a framing manipulation, analogous in spirit to the sunspot signals studied in experimental coordination games (Duffy and Fisher, 2005). In those settings, payoff-irrelevant public cues can move behavior when strategic complementarities are present, because agents may treat any salient public message as a coordination device even when it carries no information about fundamentals.

The grid, replication structure, and inference parameters are identical to Experiment 2. The only difference between the two experiments is the presence of the urgency cue in the system prompt. The identification logic is straightforward in principle. Because Experiments 2 and 3 share the same (θ, n_A^{cf}) grid, the cell-level difference

$$\Delta_{\text{panic}}(\theta, n_A^{cf}) = \Pr(\text{Stay} \mid \theta, n_A^{cf}, \text{cue}) - \Pr(\text{Stay} \mid \theta, n_A^{cf}, \text{no cue}) \quad (2)$$

measures the effect of the framing cue at each point of the policy surface. If the cue operates only through the intercept of the best-response function (raising the baseline propensity to withdraw regardless of θ or n_A^{cf}), then Δ_{panic} should be roughly uniform across the grid. If, instead, the cue changes the agent’s sensitivity to fundamentals or to coordination information, then Δ_{panic} will vary systematically across the surface, indicating interaction effects that go beyond a simple level shift.

This design connects to the theoretical distinction between fundamental-driven and panic-driven crises. Basak and Zhou (2025) define panic as withdrawal that occurs even though fundamentals would have supported the good outcome. In our setting, a uniform negative Δ_{panic} across all θ (including values where staying is payoff-dominant) would constitute direct evidence of this kind of inefficient coordination failure triggered by a non-informative cue. A more nuanced pattern, in which the cue shifts behavior primarily in the

intermediate- θ region where the decision is not dominated, would suggest that the framing effect operates through the equilibrium-selection margin rather than through a blanket increase in withdrawal propensity.

2.5 Summary of Identification Strategy

The three experiments form an escalating sequence of identification. Experiment 1 estimates the combined effect of information and coordination in a setting that mirrors standard human experiments, providing a behavioral benchmark. Experiment 2 breaks the two channels apart by exogenously fixing the coordination environment, yielding the policy surface that is the structural primitive of the theory. Experiment 3 perturbs the surface with a payoff-irrelevant cue, isolating the panic channel as a residual.

The key advantage of the LLM-agent setting is that this escalation can be implemented with exact control over the informational history. The agent specification (model, prompt structure, temperature, and output schema) is identical across experiments except for the targeted treatment variation. No learning carries over between experiments, because each decision is an independent call to the API with no memory of prior rounds or prior experiments. This is the “frozen history” property that human experiments cannot achieve: the same micro-level decision process faces different macro-level environments, and any difference in outcomes is attributable to the environment rather than to adaptation, fatigue, or carryover.

3 Empirical Results

We present the results of the three experiments in sequence. Experiment 1 validates the LLM agent against theoretical benchmarks and provides process-level evidence on deliberation. Experiment 2 maps the two-dimensional policy surface, separating the roles of fundamentals and coordination. Experiment 3 introduces a payoff-irrelevant framing cue and measures the resulting shift in the policy surface. Section 4 combines these estimates into a formal decomposition.

3.1 Experiment 1: Behavioral Validation

Each agent in Experiment 1 observes only its private hint s_i and must decide whether to withdraw, with no information about the actions of others. The experiment produces 325 independent decisions (13 values of θ , 5 seeds, 5 agents per seed).

Figure 1 (a) plots $\Pr(\text{Stay} \mid \theta)$ against the fundamental. The withdrawal curve is nearly a step function: $\Pr(\text{Stay})$ is zero for all $\theta \leq 90$, rises to 0.56 at $\theta = 100$, and reaches 0.96 at $\theta = 110$. From $\theta = 120$ onward the agent stays with certainty. This pattern is consistent with a threshold strategy of the form $a_i = \mathbf{1}\{s_i > s^*\}$, as predicted by global-games models (Carlsson and van Damme, 1993; Morris and Shin, 2003; Goldstein and Pauzner, 2005). The empirical switching point near $\theta^* \approx 100$ coincides with the break-even condition $3\theta/(1 + 2n_A) = 100$ evaluated at $n_A = 1$, suggesting that the agent’s endogenous coordination beliefs place moderate weight on one other depositor withdrawing.

Under the uniform prior and uniform noise structure of the experiment, the Bayesian posterior mean equals the signal: $E[\theta \mid s_i] = s_i$. The LLM agent’s point estimate $\hat{\theta}$ tracks the private hint almost exactly, with a correlation of 0.9998 and a mean deviation $\hat{\theta} - s_i$ of 0.13 (standard deviation 0.80). Figure 1(b) confirms this: the scatter of $\hat{\theta}$ against s_i falls tightly along the 45-degree line. $\hat{\theta}$ closely tracks the hint in this design.

The agent’s point estimate of the number of other withdrawals, $\hat{n}_A$, declines from 3.4 at $\theta = 20$ to 0.9 at $\theta = 140$ (Figure 1(c)). This gradient is qualitatively rational: a low private signal implies that other agents also receive low signals, so more of them are expected to withdraw. Across all observations, the mean $\hat{n}_A$ is 2.02, close to the unconditional expectation under symmetric beliefs.

A distinctive feature of the LLM-agent setting is that the number of reasoning tokens expended on each decision is observable. We treat this quantity as an in-silico analogue of response time, which has a long history as a process measure in experimental economics (Rubinstein, 2007; Achtziger and Alós-Ferrer, 2014). Mean token usage is 320 (standard deviation 94). Figure 1(d) and Figure 2 show that tokens vary with θ in a pattern that is consistent with decision difficulty. Near the switching region ($\theta \in \{80, 100, 110\}$), where the

expected payoff from staying is close to the safe payoff of 100, mean tokens rise to 334–348. Away from the threshold ($\theta \leq 40$ or $\theta \geq 120$), where the decision is close to dominance-solvable, tokens fall to 289–310. A similar inverted-U pattern in human response times has been documented in coordination-game experiments by [Heinemann et al. \(2009\)](#). That the LLM replicates this regularity without having been trained on response-time data provides additional process-level evidence that its decision rule responds to the degree of strategic ambiguity.

The structured-output schema requires each agent to label the dominant reason for its decision. In Experiment 1 the most frequent code is `COORDINATION_RISK_HIGH` (49 percent of responses), followed by `SIGNAL_STRONG_HIGH` (34 percent) and `SIGNAL_STRONG_LOW` (13 percent). Even without explicit coordination information, the agent’s reported reasoning is primarily about coordination risk.

3.2 Experiment 2: The Policy Surface

Experiment 2 eliminates strategic uncertainty by telling each agent with certainty that exactly n_A^{cf} of the other four depositors will choose A. This allows us to trace the full best-response surface $\Pr(\text{Stay} \mid \theta, n_A^{cf})$ over a grid of 13 values of θ and 5 levels of n_A^{cf} , producing 1,625 independent decisions.

All 1,625 agents (100 percent) report $\hat{n}_A = n_A^{cf}$, confirming that the hard-conditioning block in the prompt is fully absorbed. The dominant reason code is `KNOWN_NA_DECISIVE` (95 percent), indicating that agents correctly recognise the imposed coordination information as the primary input to the decision.

Table 1 and Figure 3 present the estimated policy surface. Three features are worth noting. First, holding n_A^{cf} fixed at any value from 1 to 4, $\Pr(\text{Stay})$ increases monotonically in θ , with a transition that is steeper at intermediate levels of n_A^{cf} . This is the information channel: changes in the private signal about fundamentals shift the withdrawal decision, net of any coordination motive. Second, holding θ fixed, $\Pr(\text{Stay})$ decreases in n_A^{cf} , consistent with the strategic complementarity in the payoff function $3\theta/(1+2n_A)$. This is the coordina-

tion channel. The sensitivity to n_A^{cf} is most pronounced at intermediate values of θ (50–90), where the decision is not dominated. Third, at $n_A^{cf} = 0$ the agent stays with probability 1.00 at every value of θ in the grid. This is a notable regularity. For $\theta \leq 33$, staying is payoff-dominated (the B payoff $3\theta < 100$), so the universal staying at $\theta = 20$ with $n_A^{cf} = 0$ represents a small rationality violation.

To summarise the surface parametrically, we fit a logit model:

$$\Pr(\text{Stay} \mid \theta, n_A^{cf}) = \Lambda\left(\beta_0 + \beta_\theta \tilde{\theta} + \beta_q \tilde{n}_A^{cf}\right), \quad (3)$$

where $\tilde{\theta}$ and $\tilde{n}_A^{cf}$ are standardised and $\Lambda(\cdot)$ is the logistic CDF. The estimates are $\hat{\beta}_0 = 2.76$ (s.e. 0.15), $\hat{\beta}_\theta = 2.16$ (0.13), and $\hat{\beta}_q = -2.51$ (0.15). The ratio $|\hat{\beta}_q/\hat{\beta}_\theta| = 1.16$ indicates that, in this environment, the coordination channel is at least as large as the information channel. Both coefficients are highly significant ($z > 16$).

3.3 Experiment 3: The Panic Channel

Experiment 3 adds a single payoff-irrelevant sentence to the prompt (“Many other depositors may act quickly”) while keeping the θ grid, hint distribution, and n_A^{cf} grid identical to Experiment 2. The resulting policy surface is reported in Table 2 and Figure 4.

The overall rate of staying falls from 75.8 percent in Experiment 2 to 52.9 percent in Experiment 3, a difference of 22.9 percentage points ($t = 14.06$, $p < 0.001$). The direction of this shift is consistent with a “panic” or sunspot effect: a payoff-irrelevant cue moves behaviour toward the run outcome. In Experiment 3, compliance with the imposed n_A^{cf} is substantially lower. While 100 percent of agents in Experiment 2 report $\hat{n}_A = n_A^{cf}$, only 34 percent do so in Experiment 3. Agents who are told with certainty that zero others will withdraw ($n_A^{cf} = 0$) report a mean $\hat{n}_A$ of 1.45. Figure 5 documents this pattern across all five levels of n_A^{cf} . The urgency cue sharply reduces compliance with the imposed n_A^{cf} ; agents behave as if the constraint were absent and instead rely on endogenous beliefs about others’ actions.

The treatment effect is far from uniform across the (θ, n_A^{cf}) grid. Table 3 and Figure 4(c)

show the cell-level difference $\Delta_{\text{panic}}(\theta, n_A^{cf}) = \Pr(\text{Stay} \mid \theta, n_A^{cf}, \text{cue}) - \Pr(\text{Stay} \mid \theta, n_A^{cf}, \text{no cue})$. The effect is largest (most negative) in the intermediate- θ range: for $\theta \in [50, 70]$, the mean cell-level shift is -0.65 , while for $\theta \in [110, 140]$ the mean shift is $+0.06$. The effect also varies with n_A^{cf} : it is strongly negative for $n_A^{cf} \in \{0, 1, 2, 3\}$ and turns positive at $n_A^{cf} = 4$. We discuss the economics behind this reversal in Section 4.

3.4 Cross-Experiment Comparison

Figure 6 plots the marginal $\Pr(\text{Stay} \mid \theta)$ curves from all three experiments on a common axis. The comparison clarifies how the three information environments produce qualitatively different withdrawal profiles.

In Experiment 1, where the agent observes only the private signal, the curve is a steep step function that transitions around $\theta = 100$. In Experiment 2, averaged over n_A^{cf} , the curve is much higher and flatter: the provision of coordination information (including $n_A^{cf} = 0$, where staying is nearly certain) pulls the overall rate of staying up to 76 percent. In Experiment 3, the urgency cue lowers the curve back toward the Experiment 1 profile, but with a less abrupt transition. The curves cross near $\theta = 90$: below that point, Experiment 3 agents stay less often than even Experiment 1 agents; above it, they stay more often.

This crossing pattern is informative. The panic cue does not simply lower the threshold. Instead, it appears to flatten the response function, making agents less sensitive to the imposed coordination condition and more reliant on their private signal about fundamentals.

4 Decomposition

We now provide a structured decomposition of the treatment effect into information, coordination, and panic components. We proceed in three steps: section 4.1 presents a pooled logit model that identifies the channels parametrically; section 4.2 describes a nonparametric description of the cell-level differences; section 4.3 provides supporting evidence from the agents' textual reasoning.

4.1 Pooled Logit with Interaction Terms

To separate the channels, we pool the observations from Experiments 2 and 3 and estimate:

$$\Pr(\text{Stay}) = \Lambda\left(\beta_0 + \beta_\theta \tilde{\theta} + \beta_q \tilde{n}_A^{cf} + \gamma_0 D + \gamma_\theta D \times \tilde{\theta} + \gamma_q D \times \tilde{n}_A^{cf}\right), \quad (4)$$

where $D = 1$ for Experiment 3 and $\tilde{\theta}$, $\tilde{n}_A^{cf}$ are standardised. The coefficients $(\beta_0, \beta_\theta, \beta_q)$ describe the baseline (Experiment 2) response, and $(\gamma_0, \gamma_\theta, \gamma_q)$ measure how the urgency cue shifts each component.

Table 4 reports the estimates. Three results stand out.

The intercept shifts strongly toward withdrawal ($\hat{\gamma}_0 = -2.29$, $z = -12.4$). In the notation of Section 2, this corresponds to the pure panic term Δ_{panic} : an increase in the baseline propensity to withdraw that is not mediated by changes in beliefs about fundamentals or others' actions. Evaluated at the means of $\tilde{\theta}$ and $\tilde{n}_A^{cf}$, the urgency cue reduces the log-odds of staying from 2.76 to 0.47. This kind of shift is what Basak and Zhou (2025) call “panic” in a formal sense: withdrawal that occurs even though fundamentals alone do not warrant it. It is also consistent with the experimental sunspot literature, where payoff-irrelevant public signals can coordinate beliefs on an inferior outcome (Cass and Shell, 1983; Duffy and Fisher, 2005).

The coefficient on θ roughly doubles ($\hat{\gamma}_\theta = +2.58$, $z = 8.7$). Under the urgency cue, agents become substantially more responsive to their private signal about fundamentals. A natural interpretation draws on the global-games logic of Morris and Shin (2003): when common knowledge about others' actions breaks down, private signals become the primary anchor for the decision, and the threshold function steepens accordingly. Heinemann et al. (2009) report a related finding in human coordination-game experiments: when strategic uncertainty is high, subjects' switching thresholds become more sensitive to fundamentals.

The sensitivity to n_A^{cf} collapses ($\hat{\gamma}_q = +2.07$, $z = 11.4$). This interaction term nearly exactly offsets the baseline coordination coefficient ($\hat{\beta}_q = -2.51$), so that the net effect of n_A^{cf} in Experiment 3 is only $-2.51 + 2.07 = -0.44$. In other words, the urgency cue reduces the responsiveness to coordination information by roughly 82 percent. This is consistent

with the conditioning compliance evidence in Figure 5: if agents do not absorb the imposed n_A^{cf} , then variation in n_A^{cf} cannot affect their decisions. The economic content of this finding is that the panic cue does not merely shift the intercept of the best-response function. It degrades the agent’s use of payoff-relevant coordination information, a mechanism that goes beyond a simple sunspot story.

4.2 Cell-Level Decomposition

The logit model in Section 4.1 imposes a parametric structure. To assess robustness, we examine the cell-level differences $\Delta(\theta, n_A^{cf})$ directly.

Averaged across the grid, the shift in $\Pr(\text{Stay})$ is -0.23 . But this average conceals large heterogeneity (Table 3). In the intermediate- θ range ($\theta \in [50, 70]$), where fundamentals are ambiguous and the decision is not dominated, the mean cell-level shift is -0.65 . This is the region where both theory and experiments predict the greatest sensitivity to coordination conditions (Goldstein and Pauzner, 2005; Heinemann et al., 2009), and it is where the urgency cue has its largest effect. At high θ ($\theta \geq 110$), the cue has little effect because the fundamentals signal is strong enough to dominate regardless of framing.

The positive differences at $n_A^{cf} = 4$ require separate discussion. In Experiment 2, an agent told that all four others will withdraw faces a B payoff of $3\theta/9 = \theta/3$. This exceeds 100 only for $\theta > 300$, well outside the support. Accordingly, agents in Experiment 2 withdraw with high probability at $n_A^{cf} = 4$ for all θ in the grid; the mean $\Pr(\text{Stay})$ at $n_A^{cf} = 4$ is 0.31. In Experiment 3, however, agents do not absorb the $n_A^{cf} = 4$ conditioning. They instead form endogenous beliefs (mean $\hat{n}_A = 1.90$ at $n_A^{cf} = 4$), which, when θ is high, lead them to stay more often than the correctly-conditioned agents in Experiment 2. The positive Δ at the south-east corner of the surface is therefore not evidence that the panic cue promotes staying. It is a mechanical consequence of the cue’s destruction of the conditioning channel, which in this region of the surface happens to move behaviour toward staying rather than away from it.

This observation connects to the broader decomposition framework. Following Kitagawa

(1955), Oaxaca (1973), and Blinder (1973), extended to nonlinear models by Fairlie (2005) and Fortin et al. (2011), we can write the total difference in mean outcomes as the sum of a composition effect (agents hold different beliefs) and a structural effect (agents respond differently to the same beliefs). In our setting, the Experiment 2 and Experiment 3 grids impose the same θ and n_A^{cf} by design, so any composition effect must come through the agents’ failure to absorb n_A^{cf} in Experiment 3. The structural effect is the change in the mapping from (θ, n_A^{cf}) to the decision, which the logit interaction terms capture. The large positive $\hat{\gamma}_q$ and the conditioning-compliance evidence together suggest that both the composition and structural channels are driven by the same underlying mechanism: the urgency cue undermines the credibility or salience of the counterfactual coordination information.

4.3 Textual Evidence

The structured-output schema requires each agent to produce a short free-text reason for its decision. We treat these reasons as corroborative evidence, with the caveat that chain-of-thought explanations need not faithfully reflect the model’s internal computations (Turpin et al., 2023).

The shift in reason codes between Experiments 2 and 3 is dramatic. In Experiment 2, 95 percent of responses are coded `KNOWN_NA_DECISIVE`. In Experiment 3, this code does not appear at all, and the leading code is `COORDINATION_RISK_HIGH` (46 percent), followed by `SIGNAL_STRONG_HIGH` (31 percent) and `RISKY_PAYOFF_ATTRACTIVE` (22 percent). Figure 8 displays this comparison. The shift is consistent with the logit evidence: when the panic cue is present, agents no longer treat the imposed n_A^{cf} as the decisive input. Instead, they reason about coordination risk and fundamentals signals in a manner qualitatively similar to Experiment 1, where no coordination information was provided.

A simple keyword search supports this reading. The word “run” (or “runs”) appears in 80 percent of Experiment 3 reasons but only 18 percent of Experiment 1 reasons. At the same time, only 0.6 percent of Experiment 3 agents at $n_A^{cf} = 0$ explicitly reference the fact that no others are withdrawing. Typical reasons at $n_A^{cf} = 0$ read: “Hint suggests moderate

θ , but payoff from B collapses if others run; 100 certain beats coordination risk.” The agent reasons as if the possibility of a run is salient and imminent, even when the prompt states with certainty that no one else is withdrawing.

4.4 Mapping the Results to the Theoretical Decomposition

We now connect the empirical evidence to the decomposition proposed in Section 2:

$$\Delta \Pr(\text{run}) = \Delta_{\text{info}} + \Delta_{\text{coord}} + \Delta_{\text{panic}} + \Delta_{\text{interaction}}. \quad (5)$$

The information component Δ_{info} is negligible. The mean $\hat{\theta}$ is 79.4 in Experiment 2 and 79.7 in Experiment 3; the difference is not significant. The urgency cue does not distort beliefs about fundamentals, consistent with its payoff-irrelevant design.

The coordination component Δ_{coord} operates through a specific channel. The mean $\hat{n}_A$ shifts from 2.00 to 1.83, a small change. But the relevant observation is not the level of coordination beliefs; it is the breakdown of the conditioning on n_A^{cf} . In Experiment 2, agents condition their decisions on the known n_A^{cf} ; in Experiment 3, they form endogenous beliefs that are largely independent of n_A^{cf} . The coordination component of the treatment effect is therefore not a simple belief shift but a structural change in how coordination information enters the decision.

The panic component Δ_{panic} corresponds to the intercept shift $\hat{\gamma}_0 = -2.29$ in the pooled logit. At the mean of the covariates, this translates to a reduction in $\Pr(\text{Stay})$ of approximately 23 percentage points. This is the component that most closely matches the “panic” definition of Basak and Zhou (2025): withdrawal in the absence of a deterioration in fundamentals.

The interaction terms are large and statistically significant ($\hat{\gamma}_\theta = +2.58$, $\hat{\gamma}_q = +2.07$). The panic channel is not additively separable from the information and coordination channels. The cue simultaneously shifts the intercept, increases signal sensitivity, and destroys coordination responsiveness. A purely additive decomposition would therefore miss the majority of the treatment effect; accounting for the interactions is essential.

In the language of global games, the urgency cue appears to “dissolve” the common-knowledge structure that the hard-conditioning block provides in Experiment 2. When common knowledge about others’ actions is present, the policy surface is relatively flat in θ (because the coordination environment is known and the decision depends primarily on whether $3\theta/(1 + 2n_A^{cf})$ exceeds 100). When common knowledge is lost, agents revert to a global-games regime in which the private signal is the dominant input and the threshold steepens. This interpretation is consistent with the theoretical results of [Angeletos and Werning \(2006\)](#) and [Angeletos et al. \(2006\)](#), who show that the introduction of public signals in global-games environments can re-introduce multiplicity and shift equilibrium selection. In our setting, the urgency cue functions as a public signal that, while uninformative about θ , effectively undermines the coordination-relevant information that was explicitly provided.

5 Discussion and Caveats

The experiments reported above suggest that LLM agents can reproduce several qualitative regularities of bank-run coordination games and that the frozen-history property of the platform permits decompositions that would be difficult to implement with human subjects. We think these results are worth developing further, but they raise concerns that deserve frank discussion.

The central maintained assumption is that LLM agents, once validated against known human regularities, can be used to explore counterfactual environments. The empirical record on this assumption is mixed. [Manning et al. \(2025\)](#) show that calibrated LLM agent samples can match human choice distributions across a range of dictator games, but uncalibrated agents often diverge substantially from human baselines. [Gao et al. \(2025\)](#) document systematic departures from human play in a simple strategic-reasoning game, and [Henning et al. \(2025\)](#) find that LLM agents in experimental asset markets converge to fundamental value far more reliably than human subjects, suppressing the bubble-and-crash dynamics routinely observed in the laboratory. For bank-run games the gap matters in a specific way: the theoretical interest of the Diamond-Dybvig environment lies in coordination failure

driven by beliefs, fear, and social contagion, and an agent that arrives at a threshold strategy by parsing a payoff table may be doing something quite different from a depositor who withdraws because her neighbour is withdrawing.

A core property of the methodology is that each decision is an independent agent with no memory of prior rounds. The claim rests in part on the assumption that setting the decoding temperature to zero produces deterministic output, so that residual variation within a cell comes from the stochastic hint draw rather than from model randomness. In practice, this assumption is only approximately true. Even at temperature zero, floating-point non-associativity in GPU arithmetic and engineering optimizations such as continuous batching can produce different outputs from identical inputs (Ouyang et al., 2025).

With those validity and implementation issues in mind, we suggest a broader discussion: the reasoning-token data suggest a direction that may prove valuable beyond the application studied here. The finding that deliberation effort peaks near the switching region echoes the inverted-U pattern in human response times documented by Rubinstein (2007) and Heinemann et al. (2009). In human experiments, measuring the cognitive process behind a strategic decision is difficult: response time is a coarse proxy, and finer-grained process data require eye-tracking equipment that is expensive, intrusive, and limited to small samples (Polonio et al., 2015; Reutskaja et al., 2011). The reasoning-token measure is cheap, available for every decision, and at least suggestive of systematic variation in computational effort across the decision surface. One natural extension would connect these data to formal models of bounded rationality. In Gabaix’s (2014) sparsity-based framework and Sims’s (2003) rational-inattention framework, cognitive effort should vary systematically with decision difficulty, and the reasoning-token data are consistent with this prediction at a qualitative level. Whether they can be disciplined tightly enough to test specific attention-allocation models is a question for future work.

6 Conclusion

This paper has proposed and implemented a methodology for using LLM agents to study bank-run coordination games. The approach is not a substitute for theory or for human experiments. It is an attempt to provide an option that trying to bridge the two methods: hold the micro-level decision process fixed while varying the macro-environment, replay identical informational histories under alternative treatments, and observe deliberation effort at the level of individual decisions.

Three findings emerge from the experiments. First, the LLM agent passes basic behavioral validation. These regularities are necessary for the methodology to be useful, but they do not imply that the agent reasons like a human depositor. Second, the policy-surface estimates from Experiment 2 reveal that the coordination channel is at least as large as the information channel. The logit coefficient on the imposed number of withdrawals is 16 percent larger in magnitude than the coefficient on fundamentals. This is consistent with the theoretical structure of the game, where strategic complementarity makes the return to staying highly sensitive to others' actions, but it is difficult to estimate cleanly in human experiments because strategic uncertainty and fundamental uncertainty are confounded. Third, the cue reduces the aggregate rate of staying and degrades the agent's responsiveness to explicitly provided coordination information. The cue appears to dissolve the common-knowledge structure that the hard-conditioning block provides, pushing agents back toward a regime in which the private signal is the dominant input. This finding connects to a theoretical literature on how public signals can reintroduce multiplicity in global-games environments ([Angeletos and Werning, 2006](#); [Angeletos et al., 2006](#)), and to experimental evidence on sunspot-driven coordination ([Duffy and Fisher, 2005](#)).

The broader point of the paper is methodological. Bank-run theory delivers comparative statics for an unrepeated game under strong epistemic assumptions. Human experiments generate data from subjects who learn and carry memories across trials. The gap between these two modes of inquiry is about whether the theoretical object and the experimental object are the same thing. LLM agents do not close this gap on their own. But they expand

the set of feasible identification moves: we can now run the counterfactual exercises that theory assumes and that human experiments approximate, in a setting where the history is literally frozen and the only thing that changes is the treatment. Whether this capability proves useful beyond the bank-run application studied here is an empirical question. The platform is general enough to accommodate other coordination games, network structures, and policy interventions, and we regard extensions along these dimensions as natural next steps.

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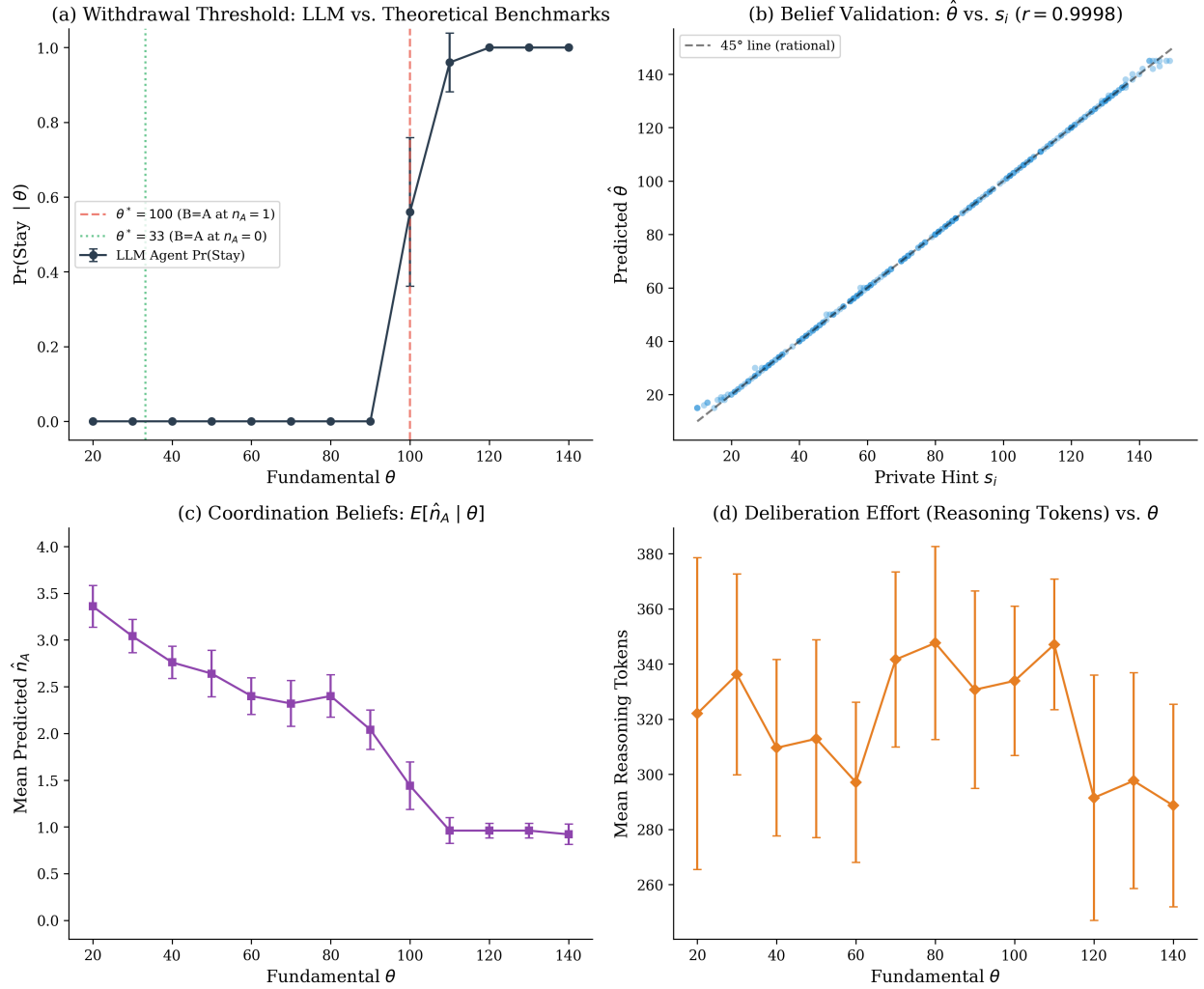


Figure 1: **Behavioral validation from Experiment 1.**

This figure reports behavioral diagnostics for Experiment 1, in which each agent observes only its private hint s_i with no information about others' actions. Panel (a) plots the fraction of agents choosing B (stay) as a function of θ , with 95% confidence intervals; dashed vertical lines mark the theoretical break-even values of θ at $n_A = 0$ and $n_A = 1$. Panel (b) plots each agent's point estimate $\hat{\theta}$ against its private hint s_i ; the dashed line is the 45-degree Bayesian benchmark. Panel (c) reports the mean predicted number of other withdrawals $\hat{n}_A$ as a function of θ . Panel (d) reports mean reasoning tokens, a proxy for deliberation effort, as a function of θ . The experiment sweeps θ over the grid $\{20, 30, \dots, 140\}$ (13 grid points). Each grid point comprises 25 independent decisions (5 seeds $\times$ 5 agents). All error bars are ± 1.96 standard errors.

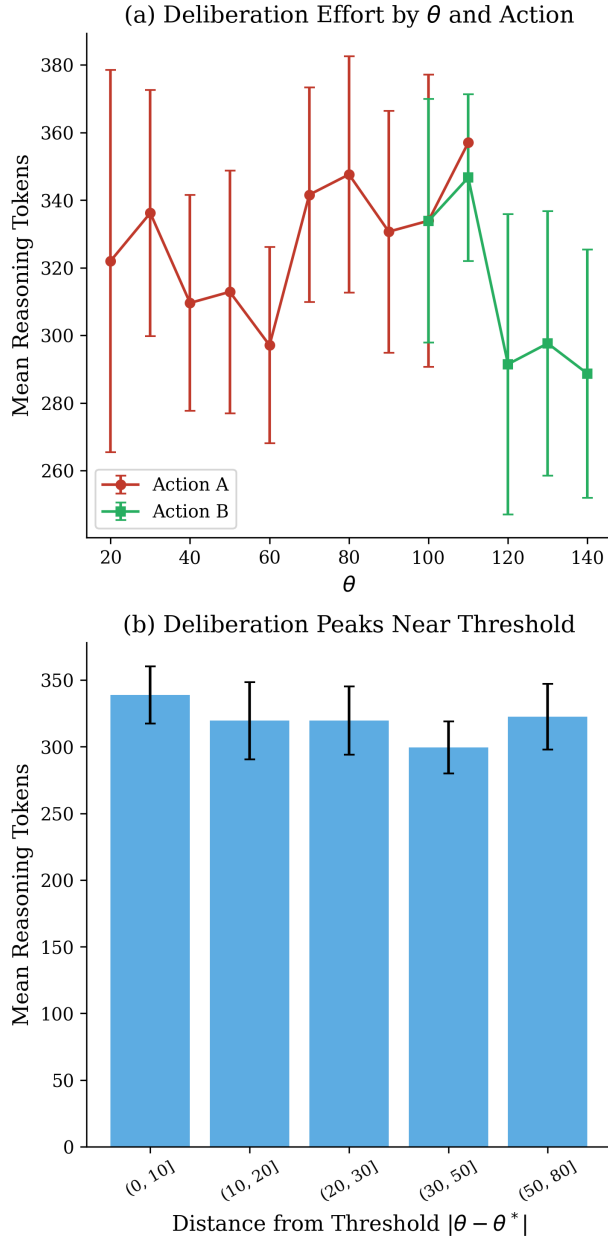


Figure 2: **Deliberation effort in Experiment 1.**

This figure examines the relationship between reasoning-token usage and decision environment in Experiment 1. Panel (a) reports mean reasoning tokens by θ , separately for agents who chose A (withdraw) and B (stay). Panel (b) plots mean reasoning tokens as a function of the absolute distance from the empirical switching point. The sample covers 3,025 independent decisions across 121 values of θ , with 25 decisions per grid point. Error bars are ± 1.96 standard errors.

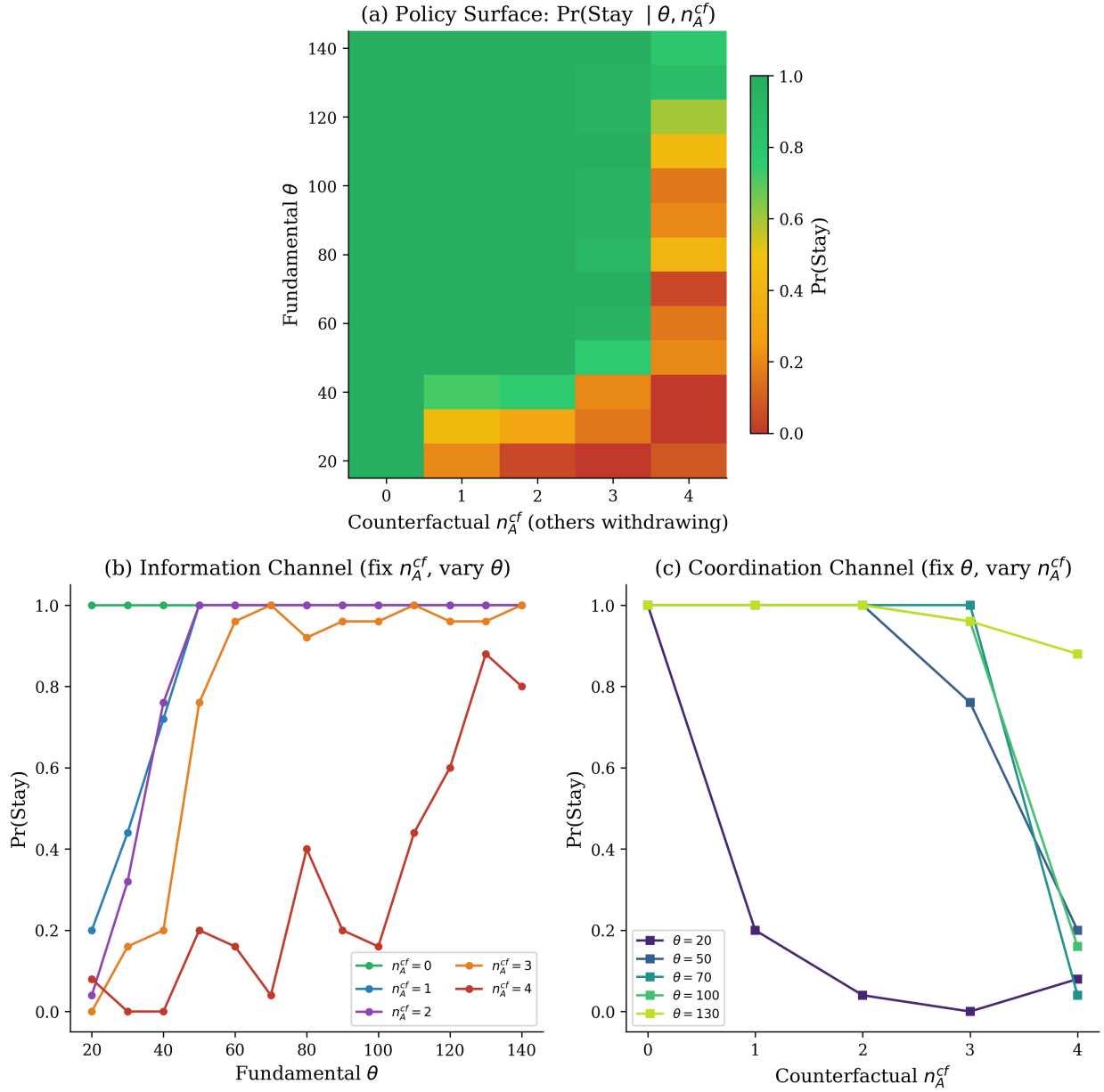


Figure 3: **Estimated best-response surface from Experiment 2.**

This figure presents the empirical best-response surface $\Pr(\text{Stay} \mid \theta, n_A^{cf})$ estimated in Experiment 2, where each agent is told with certainty that exactly n_A^{cf} of the other four depositors withdraw. Panel (a) displays a heat map of $\Pr(\text{Stay})$ over the (θ, n_A^{cf}) grid. Panel (b) plots $\Pr(\text{Stay})$ as a function of θ with n_A^{cf} held fixed at each level (information channel). Panel (c) plots $\Pr(\text{Stay})$ as a function of n_A^{cf} with θ held fixed at selected values (coordination channel). The grid covers $\theta \in \{20, 30, \dots, 140\}$ (13 values) crossed with $n_A^{cf} \in \{0, 1, 2, 3, 4\}$ (5 levels), producing 65 cells. Each cell contains 25 independent decisions (5 seeds $\times$ 5 agents), for a total of 1,625 observations.

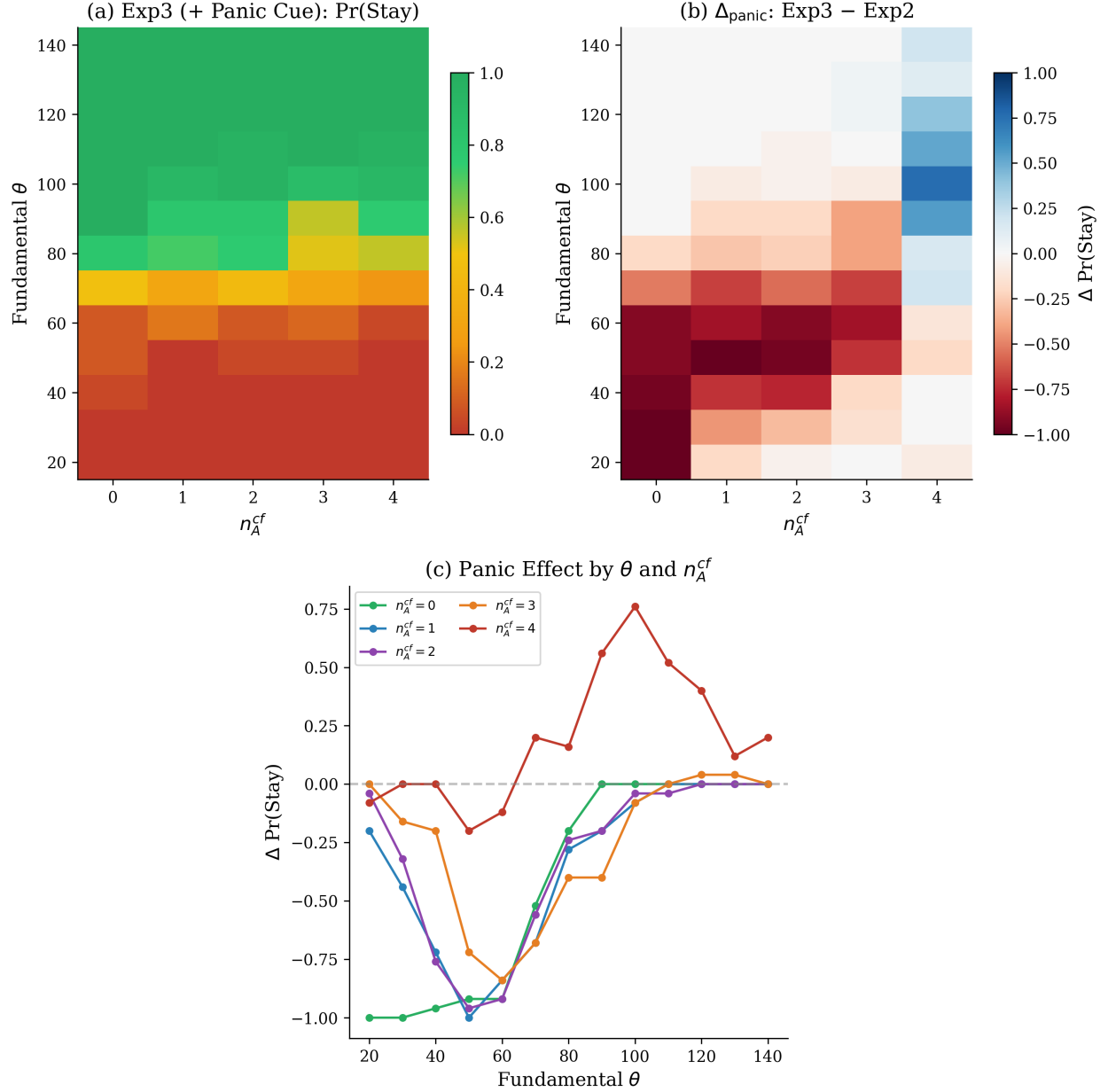


Figure 4: **Panic channel on the best-response surface.**

This figure compares the best-response surfaces estimated in Experiments 2 and 3. In Experiment 3 the system prompt includes a single payoff-irrelevant urgency cue; all other design elements are identical to Experiment 2. Panel (a) presents the Experiment 3 policy surface $\text{Pr}(\text{Stay} \mid \theta, n_A^{cf})$. Panel (b) displays the cell-level difference $\Delta_{\text{panic}} = \text{Pr}(\text{Stay}, \text{Exp3}) - \text{Pr}(\text{Stay}, \text{Exp2})$; blue shading indicates higher staying rates under the cue and red shading indicates higher withdrawal rates. Panel (c) plots Δ_{panic} as a function of θ , separately for each level of n_A^{cf} . Both experiments use the same 13×5 grid, with 25 independent decisions per cell.

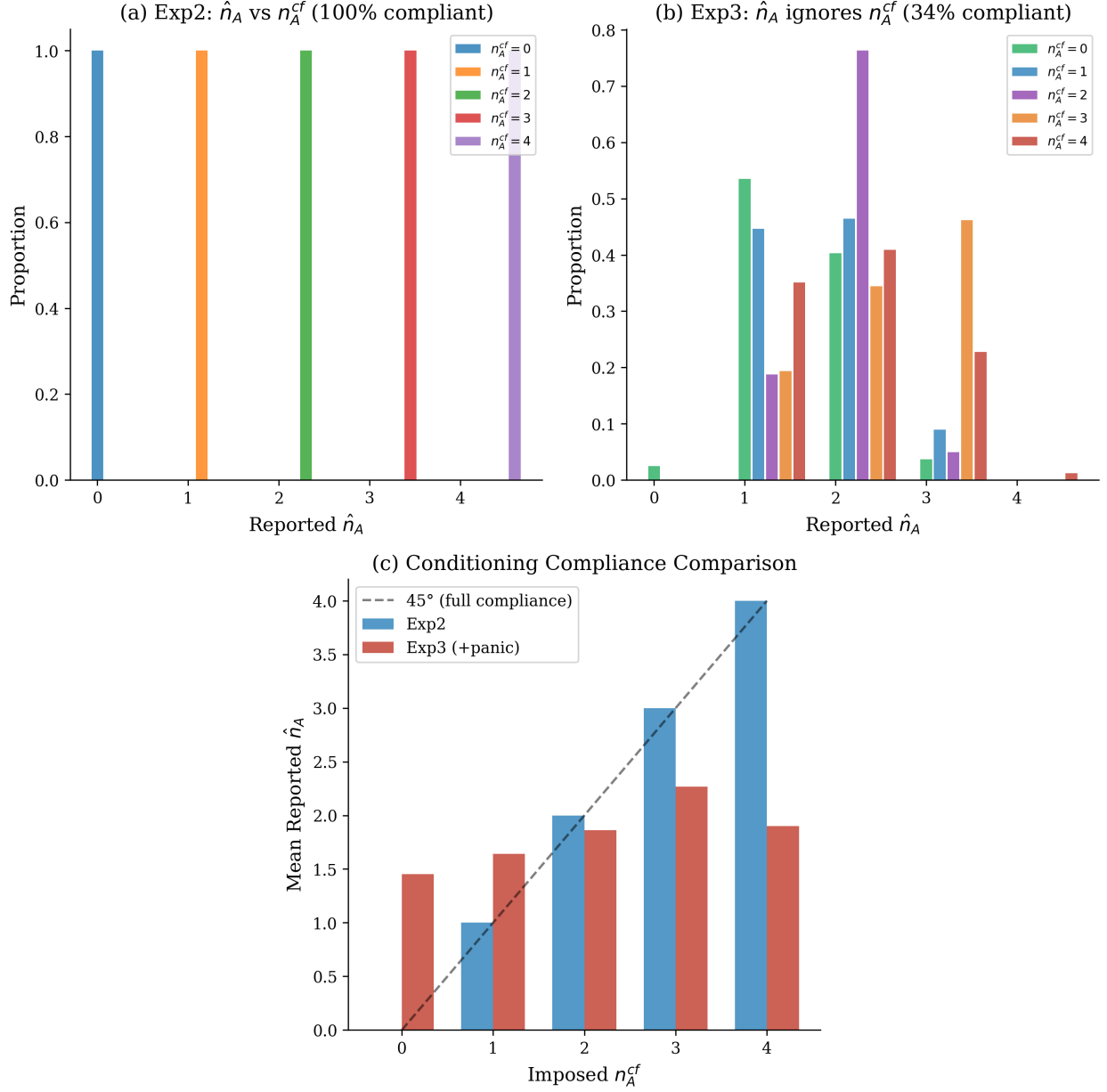


Figure 5: **Conditioning compliance across Experiments 2 and 3.**

This figure examines whether agents absorb the hard-conditioning block by comparing each agent's reported belief $\hat{n}_A$ against the imposed counterfactual value n_A^{cf} . Panel (a) displays the joint distribution of reported $\hat{n}_A$ and imposed n_A^{cf} in Experiment 2. Panel (b) displays the same joint distribution in Experiment 3, where the system prompt includes the urgency cue. Panel (c) plots mean reported $\hat{n}_A$ against the imposed n_A^{cf} for both experiments; the 45-degree line marks full compliance. Each experiment contains 1,625 observations (65 cells $\times$ 25 decisions per cell).

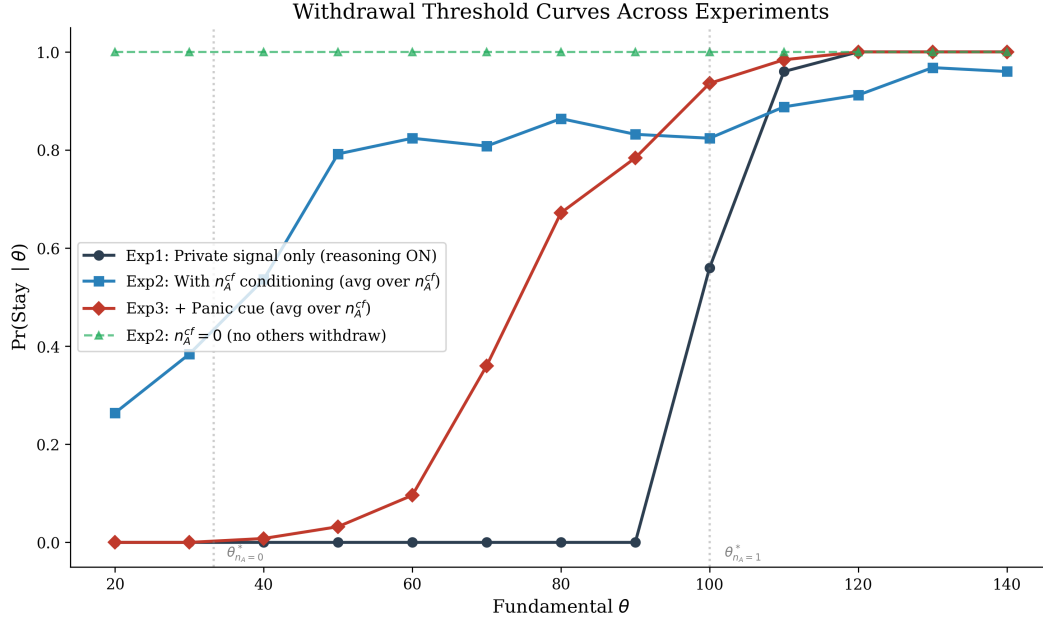


Figure 6: **Comparison of withdrawal threshold curves across all three experiments.**

This figure plots $\Pr(\text{Stay} \mid \theta)$ on a common axis for all three experiments. For Experiment 1, the curve is computed from the private-signal-only condition (3,025 decisions across 121 values of θ). For Experiments 2 and 3, the curve is averaged over all five levels of n_A^{cf} (1,625 decisions each across 13 values of θ). Experiment 1 provides only a private hint; Experiment 2 adds hard-conditioning on n_A^{cf} ; Experiment 3 adds both the n_A^{cf} grid and the payoff-irrelevant urgency cue.

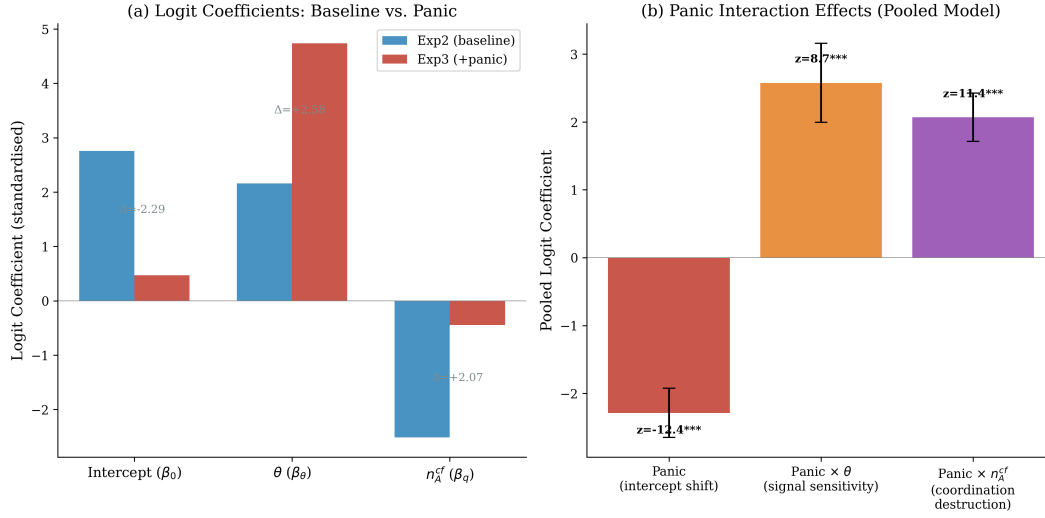


Figure 7: **Logit decomposition of the panic channel.**

This figure summarizes the pooled logit model in equation (4), estimated on the combined sample of Experiments 2 and 3 ($N = 3,250$). Panel (a) compares the three baseline logit coefficients (β_0 , β_θ , β_q) between Experiment 2 and Experiment 3. Panel (b) displays the three interaction terms (γ_0 , γ_θ , γ_q) that measure how the urgency cue shifts the intercept, the sensitivity to fundamentals, and the sensitivity to coordination information, respectively. All regressors are standardized to zero mean and unit variance. Error bars represent 95% confidence intervals.

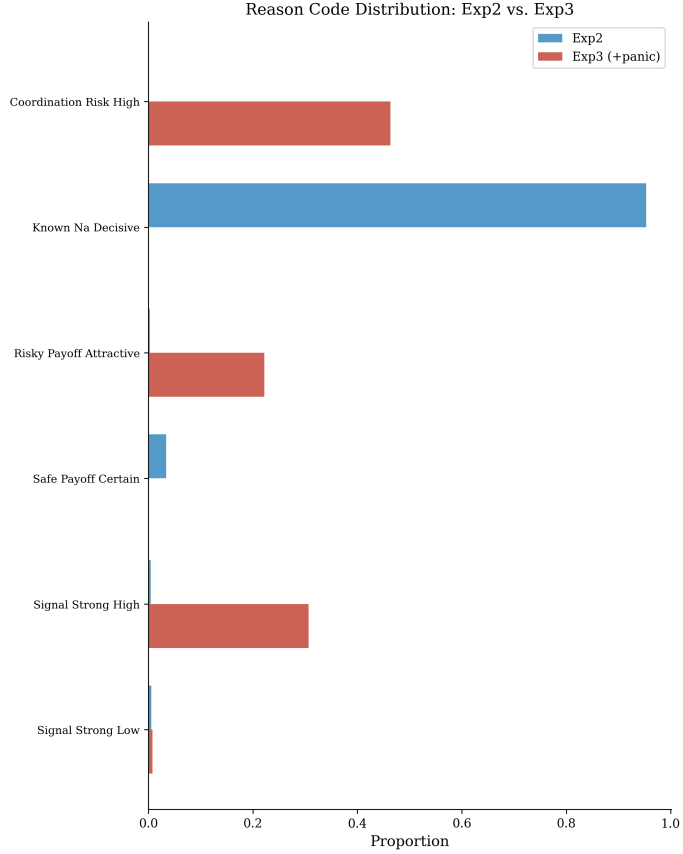


Figure 8: **Shift in reason-code distributions between Experiments 2 and 3.**

This figure compares the distribution of agent-reported reason codes across Experiments 2 and 3. Each agent selects a categorical reason code from a fixed enumeration as part of the structured-output schema. The bars report the fraction of agents in each experiment selecting each reason code. Experiment 2 contains 1,625 observations (with hard-conditioning on n_A^{cf}); Experiment 3 contains 1,625 observations (with the payoff-irrelevant urgency cue). Reason codes are drawn from the enumeration defined in the structured-output schema (Appendix A.1.3).

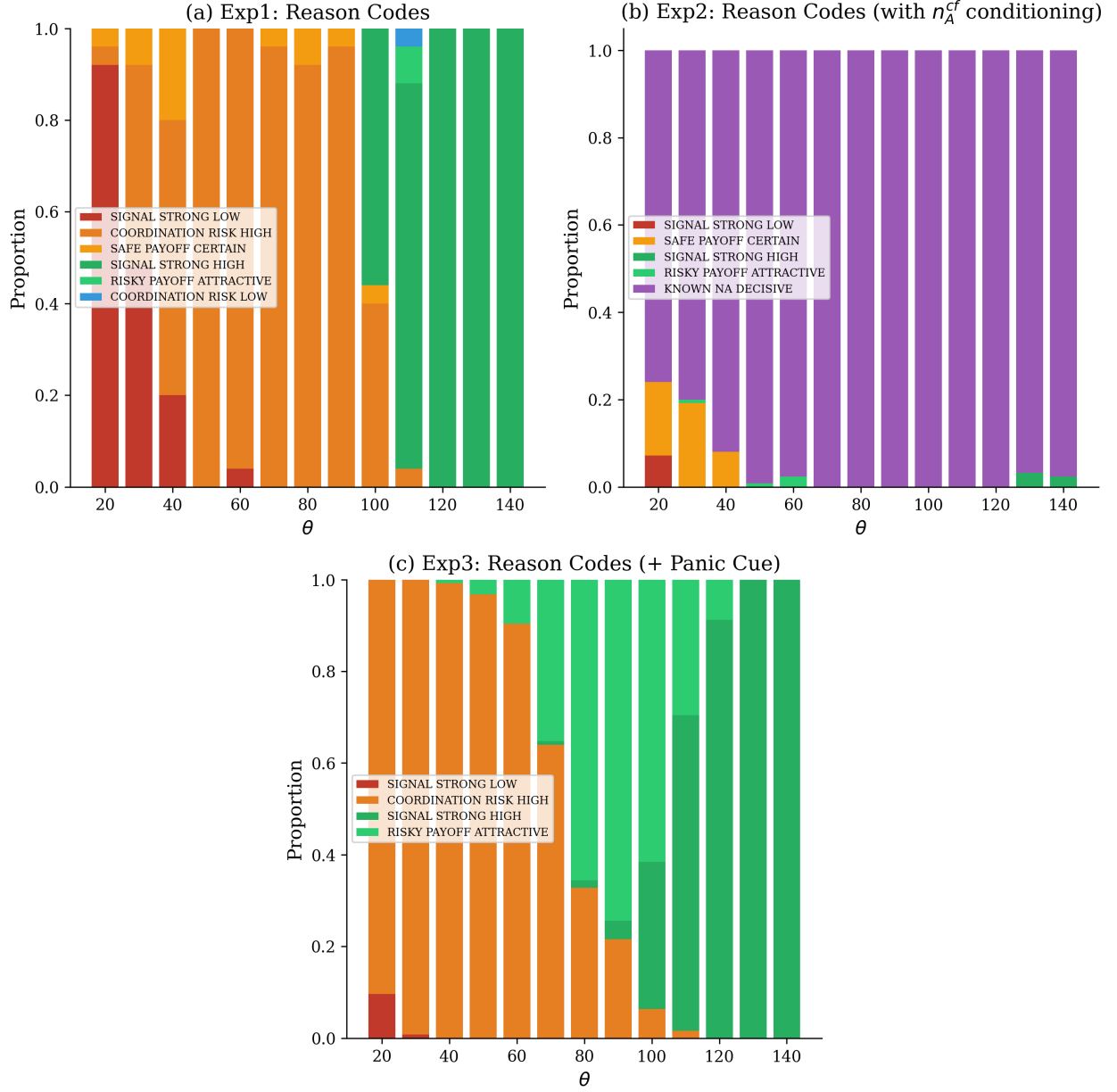


Figure 9: **Distribution of reason codes by θ for each experiment.**

This figure displays the composition of agent-reported reason codes as a function of θ for each experiment. Panel (a) shows Experiment 1 (private signal only; 25 decisions per θ , 121 grid points). Panel (b) shows Experiment 2 (with hard-conditioning on n_A^{cf} ; 25 decisions per cell, pooled across 5 levels of n_A^{cf}). Panel (c) shows Experiment 3 (with the payoff-irrelevant urgency cue; same grid and replication structure as Experiment 2). Reason codes are drawn from the enumeration defined in the structured-output schema (Appendix A.1.3).

Table 1. **Estimated best-response surface from Experiment 2.**

This table reports the empirical best-response surface $\Pr(\text{Stay} \mid \theta, n_A^{cf})$ from Experiment 2, in which each agent is told with certainty that exactly n_A^{cf} of the other four depositors will choose Action A. Rows index the fundamental $\theta \in \{20, 30, \dots, 140\}$ and columns index the imposed number of other withdrawals $n_A^{cf} \in \{0, 1, 2, 3, 4\}$. Each cell reports the fraction of 25 independent decisions (5 seeds $\times$ 5 agents) in which the agent chooses Action B (stay). The reasoning-effort parameter is set to “none” and temperature to zero, so that within-cell variation reflects only the stochastic hint draw.

θ	n_A^{cf}				
	0	1	2	3	4
20	1.00	0.20	0.04	0.00	0.08
30	1.00	0.44	0.32	0.16	0.00
40	1.00	0.72	0.76	0.20	0.00
50	1.00	1.00	1.00	0.76	0.20
60	1.00	1.00	1.00	0.96	0.16
70	1.00	1.00	1.00	1.00	0.04
80	1.00	1.00	1.00	0.92	0.40
90	1.00	1.00	1.00	0.96	0.20
100	1.00	1.00	1.00	0.96	0.16
110	1.00	1.00	1.00	1.00	0.44
120	1.00	1.00	1.00	0.96	0.60
130	1.00	1.00	1.00	0.96	0.88
140	1.00	1.00	1.00	1.00	0.80

Table 2. **Estimated best-response surface from Experiment 3.**

This table reports the empirical best-response surface $\Pr(\text{Stay} \mid \theta, n_A^{cf})$ from Experiment 3. The experimental setup is identical to Experiment 2 (Table 1) except that the system prompt includes a payoff-irrelevant urgency cue (“Many other depositors may act quickly”). Rows index the fundamental $\theta \in \{20, 30, \dots, 140\}$ and columns index the imposed number of other withdrawals $n_A^{cf} \in \{0, 1, 2, 3, 4\}$. Each cell reports the fraction of 25 independent decisions (5 seeds $\times$ 5 agents) in which the agent chooses Action B (stay). The grid, replication structure, hint distribution, and inference parameters are identical to Experiment 2.

θ	n_A^{cf}				
	0	1	2	3	4
20	0.00	0.00	0.00	0.00	0.00
30	0.00	0.00	0.00	0.00	0.00
40	0.04	0.00	0.00	0.00	0.00
50	0.08	0.00	0.04	0.04	0.00
60	0.08	0.16	0.08	0.12	0.04
70	0.48	0.32	0.44	0.32	0.24
80	0.80	0.72	0.76	0.52	0.56
90	1.00	0.80	0.80	0.56	0.76
100	1.00	0.92	0.96	0.88	0.92
110	1.00	1.00	0.96	1.00	0.96
120	1.00	1.00	1.00	1.00	1.00
130	1.00	1.00	1.00	1.00	1.00
140	1.00	1.00	1.00	1.00	1.00

Table 3. **Cell-level panic effect: difference in $\Pr(\text{Stay})$ between Experiments 3 and 2.**

This table reports the cell-level treatment effect of the payoff-irrelevant urgency cue. Each cell is computed as $\Delta_{\text{panic}}(\theta, n_A^{cf}) = \Pr(\text{Stay}, \text{Exp 3}) - \Pr(\text{Stay}, \text{Exp 2})$, where the constituent probabilities are taken from Tables 2 and 1, respectively. Rows index the fundamental $\theta \in \{20, 30, \dots, 140\}$ and columns index the imposed number of other withdrawals $n_A^{cf} \in \{0, 1, 2, 3, 4\}$. Negative values indicate that the urgency cue shifts behaviour toward withdrawal; positive values indicate the opposite. Each underlying cell probability is based on 25 independent decisions.

θ	n_A^{cf}				
	0	1	2	3	4
20	-1.00	-0.20	-0.04	0.00	-0.08
30	-1.00	-0.44	-0.32	-0.16	0.00
40	-0.96	-0.72	-0.76	-0.20	0.00
50	-0.92	-1.00	-0.96	-0.72	-0.20
60	-0.92	-0.84	-0.92	-0.84	-0.12
70	-0.52	-0.68	-0.56	-0.68	+0.20
80	-0.20	-0.28	-0.24	-0.40	+0.16
90	0.00	-0.20	-0.20	-0.40	+0.56
100	0.00	-0.08	-0.04	-0.08	+0.76
110	0.00	0.00	-0.04	0.00	+0.52
120	0.00	0.00	0.00	+0.04	+0.40
130	0.00	0.00	0.00	+0.04	+0.12
140	0.00	0.00	0.00	0.00	+0.20

Table 4. **Pooled logit estimates for the panic-channel decomposition.**

This table reports maximum-likelihood estimates of the pooled logit model in equation (4). The dependent variable equals one if the agent chooses Action B (stay) and zero if it chooses Action A (withdraw). The baseline coefficients ($\beta_0, \beta_\theta, \beta_q$) capture the Experiment 2 response surface; the interaction terms ($\gamma_0, \gamma_\theta, \gamma_q$) measure how the urgency cue in Experiment 3 shifts each component. The indicator D equals one for Experiment 3 and zero for Experiment 2. All continuous regressors ($\tilde{\theta}, \tilde{n}_A^{cf}$) are standardised to have zero mean and unit variance over the pooled sample. The sample pools all observations from Experiments 2 and 3 ($N = 3,250$; 25 decisions per cell $\times$ 65 cells $\times$ 2 experiments). Standard errors are reported in parentheses. All coefficients are significant at the 1% level ($|z| > 8$ in every case).

	Coefficient	Std. Error
Intercept (β_0)	2.759	(0.152)
$\tilde{\theta}$ (β_θ)	2.157	(0.134)
$\tilde{n}_A^{cf}$ (β_q)	-2.512	(0.149)
D (γ_0)	-2.288	(0.184)
$D \times \tilde{\theta}$ (γ_θ)	2.576	(0.297)
$D \times \tilde{n}_A^{cf}$ (γ_q)	2.068	(0.182)
Log-likelihood	-742.8	

Appendix

A Experimental Methodology and Prompt Design

This appendix provides a description of the three experiments that constitute the empirical backbone of the paper. For each experiment we document (i) the economic environment presented to the LLM agent, (ii) the exact prompt text, (iii) the parameterisation grid, and (iv) the structured-output schema that enforces machine-readable responses. All experiments share the same underlying bank-run game, a five-depositor coordination game following [Bayona and Peia \(2022\)](#), but differ in the information and framing conditions imposed on each agent.

A.1 Common Environment

A.1.1 The Bank-Run Game

A group of $N = 5$ depositors simultaneously choose between two actions.

Action A (withdraw). The depositor receives a safe payoff of 100 points with certainty.

Action B (stay). The depositor receives a risky payoff that depends on an unobserved fundamental θ and on the number of other depositors who choose A:

$$\pi_B(\theta, n_A) = \frac{3\theta}{1 + 2n_A}, \quad (6)$$

where $n_A \in \{0, 1, 2, 3, 4\}$ denotes the number of the other four depositors who withdraw.

A.1.2 Information Structure

The true state of the economy θ is drawn uniformly from the integer-valued support $[\theta_{\min}, \theta_{\max}] = [10, 150]$. Each depositor i receives a private hint s_i drawn uniformly from $[\theta - 10, \theta + 10]$, clipped to $[\theta_{\min}, \theta_{\max}]$. That is, the half-width of the signal noise is $\epsilon = 10$. The signal structure is common knowledge among agents.

A.1.3 Structured-Output Schema

Every LLM call returns a JSON object conforming to a Pydantic `AgentDecision` schema with the following fields:

Field	Type	Description
<code>action</code>	<code>{"A", "B"}</code>	The depositor’s withdrawal decision
<code>predicted_theta</code>	integer in $[10, 150]$	Point estimate of θ
<code>predicted_nA</code>	integer in $[0, 4]$	Point estimate of the number of other withdrawals
<code>reason_code</code>	enumerated string	Categorical label for the dominant reason
<code>reason</code>	string (≤ 160 chars)	Free-text explanation

The `reason_code` field takes values from: `SAFE_PAYOFF_CERTAIN`, `RISKY_PAYOFF_ATTRACTIVE`, `COORDINATION_RISK_HIGH`, `COORDINATION_RISK_LOW`, `SIGNAL_STRONG_HIGH`, `SIGNAL_STRONG_LOW`, `UNCERTAIN`, `OTHER`. In Experiments 2 and 3, which include counterfactual n_A conditioning, the additional code `KNOWN_NA_DECISIVE` is available.

This schema is enforced via OpenAI’s Structured Outputs API (`response_format = AgentDecision`), guaranteeing that every successful call returns machine-parseable data without post-hoc extraction.

A.1.4 LLM Configuration and Reproducibility

All experiments use GPT-5.2 accessed through Azure OpenAI. We use GPT-5.2 with setting temperature as 0.

A.2 Experiment 1: Calibration and Behavioral Validation

A.2.1 Design

- **Grid:** $\theta \in \{20, \dots, 140\}$ (13 grid points).
- **Replication:** 5 seeds $\times$ 5 agents = 25 independent decisions per θ .
- **Total calls:** $13 \times 25 = 325$.
- **Information condition:** Each agent observes only its private hint s_i . No information about the number of other withdrawals is provided.

A.2.2 System Prompt

The system prompt places the agent in the role of a bank depositor and communicates the rules of the game:

You are a depositor in a bank. You must decide now:

You are in a group of 5 investors. Everyone chooses at the same time.

Your choice

- Choose A: You obtain back your initial investment now (**100** points for sure).
- Choose B: You keep your position and receive a payoff shown in the table below.

Payoffs

- If you choose A: $\text{payoff} = 100$ points for sure.
- If you choose B: $\text{payoff} = 3 \times (\text{State of the economy}) / (1 + 2 \times n_A)$, where n_A is the number of OTHER investors in your group who choose Action A.

Information

The true State of the economy is a number between 10 and 150 (all values are equally likely). You do NOT observe it directly. Instead, you receive a private hint. The hint is randomly drawn from $[\text{State} - 10, \text{State} + 10]$. All numbers in that interval are equally likely. Other investors receive their own hints.

Example Payoff table for choosing B

# others choosing A	$\theta=10$	$\theta=25$	$\theta=50$	$\theta=75$	$\theta=100$	$\theta=125$	$\theta=150$
0	30	75	150	225	300	375	450
1	10	25	50	75	100	125	150
2	6	15	30	45	60	75	90
3	4.3	10.7	21.4	32.1	42.9	53.6	64.3
4	3.3	8.3	16.7	25.0	33.3	41.7	50.0

Task

Step 1) Make your choice FIRST (A or B). Do not show calculations.

Step 2) AFTER you have chosen, give your best guesses about θ and n_A as a quick, intuitive forecast.

- Do NOT revise your choice after making the forecasts.
- The forecasts are only for reporting; they do not need to be exact.

Return ONLY valid JSON matching the schema.

A.2.3 User Prompt

For each call the user prompt communicates the realised hint:

Your private hint about the State of the economy is: $\{s_i\}$

Now choose Action A or Action B.

Remember: n_A counts ONLY OTHER investors (not you).

Return JSON only.

A.3 Experiment 2: Policy Surface with Counterfactual Coordination Information

A.3.1 Design

- **Grid:** $\theta \in \{20, 30, \dots, 140\}$ (13 grid points) $\times n_A^{cf} \in \{0, 1, 2, 3, 4\}$ (5 levels).
- **Replication:** 5 seeds $\times$ 5 agents = 25 per cell.
- **Total cells:** $13 \times 5 = 65$; total calls: $65 \times 25 = 1,625$.
- **Information condition:** Each agent observes its private hint and is told with certainty that exactly n_A^{cf} of the other four investors will choose A.

A.3.2 System Prompt

The system prompt is identical to Experiment 1 with one critical addition: a hard-conditioning block that removes strategic uncertainty:

What you know about other investors

You are 100% sure that exactly $\{n_A^{cf}\}$ of the OTHER investors will choose Action A.

This is certain information, not a guess.

The full prompt is:

You are a depositor in a bank. You must decide now:

You are in a group of 5 investors. Everyone chooses at the same time.

Your choice

- *Choose A: You obtain back your initial investment now (**100** points for sure).*
- *Choose B: You keep your position and receive a payoff shown in the table below.*

Payoffs

- *If you choose A: payoff = 100 points for sure.*
- *If you choose B: payoff = $3 \times (\text{State of the economy}) / (1 + 2 \times n_A)$, where n_A is the number of OTHER investors in your group who choose Action A.*

Information

The true State of the economy is a number between 10 and 150 (all values are equally

likely). You do NOT observe it directly. Instead, you receive a private hint. The hint is randomly drawn from $[State - 10, State + 10]$. All numbers in that interval are equally likely. Other investors receive their own hints.

Example Payoff table for choosing B

# others choosing A	$\theta=10$	$\theta=25$	$\theta=50$	$\theta=75$	$\theta=100$	$\theta=125$	$\theta=150$
0	30	75	150	225	300	375	450
1	10	25	50	75	100	125	150
2	6	15	30	45	60	75	90
3	4.3	10.7	21.4	32.1	42.9	53.6	64.3
4	3.3	8.3	16.7	25.0	33.3	41.7	50.0

What you know about other investors

You are 100% sure that exactly n_A of the OTHER investors will choose Action A. This is certain information, not a guess.

Task

Step 1) Make your choice FIRST (A or B). Do not show calculations.

Step 2) AFTER you have chosen, give your best guesses about θ and n_A as a quick, intuitive forecast.

- Do NOT revise your choice after making the forecasts.
- The forecasts are only for reporting; they do not need to be exact.

A.3.3 User Prompt

The user prompt is identical in structure to Experiment 1:

Your private hint about the State of the economy is: $\{s_i\}$

Now choose Action A or Action B.

Remember: n_A counts ONLY OTHER investors (not you).

Return JSON only.

A.4 Experiment 3: Urgency Framing and the Panic Channel

A.4.1 Design

- **Grid:** Identical to Experiment 2: $\theta \in \{20, 30, \dots, 140\} \times n_A^{cf} \in \{0, 1, 2, 3, 4\}$.
- **Replication:** 5 seeds $\times$ 5 agents = 25 per cell.
- **Total calls:** 1,625.
- **Treatment:** The system prompt appends a payoff-irrelevant public cue (a “sunspot”-

style statement) immediately before the task instructions: “*Many other depositors may act quickly.*”

A.4.2 System Prompt

The system prompt is identical to Experiment 2 except for the insertion of the following sentence after the payoff table and before the task block:

Many other depositors may act quickly.

This sentence is intended to be payoff-irrelevant in the sense that it does not alter the payoffs, the distribution of θ or hints, or the imposed counterfactual n_A^{cf} . It is a pure framing manipulation. While it does not provide precise, verifiable information about fundamentals, it resembles the kind of extrinsic public cue discussed in “sunspot” narratives and in experimental discussions of panic-like episodes, where behavior can shift because agents treat a salient message as relevant for coordination.

The full prompt is:

You are a depositor in a bank. You must decide now:

You are in a group of 5 investors. Everyone chooses at the same time.

Your choice

- *Choose A: You obtain back your initial investment now (**100** points for sure).*
- *Choose B: You keep your position and receive a payoff shown in the table below.*

Payoffs

- *If you choose A: payoff = 100 points for sure.*
- *If you choose B: payoff = $3 \times (\text{State of the economy}) / (1 + 2 \times n_A)$, where n_A is the number of OTHER investors in your group who choose Action A.*

Information

The true State of the economy is a number between 10 and 150 (all values are equally likely). You do NOT observe it directly. Instead, you receive a private hint. The hint is randomly drawn from $[\text{State} - 10, \text{State} + 10]$. All numbers in that interval are equally likely. Other investors receive their own hints.

Example Payoff table for choosing B

<i># others choosing A</i>	$\theta=10$	$\theta=25$	$\theta=50$	$\theta=75$	$\theta=100$	$\theta=125$	$\theta=150$
0	30	75	150	225	300	375	450
1	10	25	50	75	100	125	150
2	6	15	30	45	60	75	90
3	4.3	10.7	21.4	32.1	42.9	53.6	64.3
4	3.3	8.3	16.7	25.0	33.3	41.7	50.0

What you know about other investors

You are 100% sure that exactly n_A of the OTHER investors will choose Action A. This is certain information, not a guess.

Many other depositors may act quickly.

Task

Step 1) Make your choice FIRST (A or B). Do not show calculations.

Step 2) AFTER you have chosen, give your best guesses about θ and n_A as a quick, intuitive forecast.

- *Do NOT revise your choice after making the forecasts.*
- *The forecasts are only for reporting; they do not need to be exact.*

A.4.3 User Prompt

Identical to Experiments 1 and 2.